Supplementary Material: CRBench
A Method-Agnostic, Resource-Aware Evaluation Framework for Long-Context Large Language Models

Anonymous authors
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A Full Per-Model Leaderboards

This section provides complete Part 1 (Algorithmic Representation Fidelity $\mathcal{S}_{\text{res}}$) and Part 2 (Hardware System Serving Utility $\mathcal{S}_{\text{sys}}$) leaderboards for all three evaluated foundation model families across 3,150 queries.

A.1 Gemma 4 E2B Leaderboard (2K $\rightarrow$ 128K Context Ladder, 1,050 Queries)

Table 1: Gemma 4 E2B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**SnapKV**	46.8%	6.00 bpt	62.5%	**51.5**
2	**KIVI**	46.4%	6.01 bpt	62.4%	51.2
3	**DKV-Mid**	44.7%	5.46 bpt	65.9%	51.0
4	**DKV-High**	46.7%	10.78 bpt	32.6%	42.5
5	**Low-Rank KV**	31.2%	6.07 bpt	62.1%	40.5
6	**StreamingLLM**	22.9%	6.00 bpt	62.5%	34.8
7	**Dense FP16** (Reference)	46.4%	16.00 bpt	0.0%	32.5
8	**KV-Merging**	13.3%	6.01 bpt	62.5%	28.0

Table 2: Gemma 4 E2B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KIVI**	**64.6**	51.2	12.51s	11.9 tok/s	847.0 MB	0.96×
2	**SnapKV**	60.6	51.5	13.04s	12.9 tok/s	2088.2 MB	0.82×
3	**DKV-Mid**	59.1	51.0	22.26s	13.8 tok/s	1389.9 MB	0.78×
4	**Low-Rank KV**	57.4	40.5	12.80s	12.3 tok/s	849.8 MB	0.97×
5	**StreamingLLM**	54.3	34.8	12.58s	13.7 tok/s	847.4 MB	1.00×
6	**Dense FP16**	52.8	32.5	13.18s	13.4 tok/s	1118.4 MB	1.00×
7	**DKV-High**	52.7	42.5	23.40s	13.8 tok/s	1391.0 MB	0.77×
8	**KV-Merging**	49.6	28.0	12.59s	13.7 tok/s	847.8 MB	1.00×

A.2 Gemma 4 E4B Leaderboard (2K → 32K Context Ladder, 1,050 Queries)

Table 3: Gemma 4 E4B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**KIVI**	62.0%	6.04 bpt	62.2%	**62.1**
2	**SnapKV**	60.0%	6.00 bpt	62.5%	60.8
3	**DKV-Mid**	55.3%	5.00 bpt	68.7%	59.4
4	**DKV-High**	61.3%	10.68 bpt	33.3%	52.9
5	**Low-Rank KV**	48.8%	6.27 bpt	60.8%	52.4
6	**Dense FP16** (Reference)	64.0%	16.00 bpt	0.0%	44.8
7	**StreamingLLM**	25.5%	6.00 bpt	62.5%	36.6
8	**KV-Merging**	25.3%	6.03 bpt	62.3%	36.4

Table 4: Gemma 4 E4B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KIVI**	**73.4**	62.1	2.84s	12.8 tok/s	761.3 MB	1.00×
2	**DKV-Mid**	66.3	59.4	5.06s	12.3 tok/s	760.7 MB	0.83×
3	**Low-Rank KV**	65.9	52.4	3.18s	13.1 tok/s	761.5 MB	0.97×
4	**Dense FP16**	61.4	44.8	2.96s	12.4 tok/s	761.5 MB	1.00×
5	**DKV-High**	60.0	52.9	6.25s	12.2 tok/s	761.0 MB	0.76×
6	**StreamingLLM**	55.6	36.6	2.92s	13.6 tok/s	761.4 MB	1.00×
7	**KV-Merging**	55.5	36.4	2.91s	13.5 tok/s	761.5 MB	1.00×
8	**SnapKV**	48.6	60.8	83.38s	10.6 tok/s	2632.4 MB	0.20×

A.3 Qwen2.5 7B Leaderboard ($2\text{K} \rightarrow 32\text{K}$ Context Ladder, 1,050 Queries)

Table 5: Qwen2.5 7B Part 1 ($\mathcal{S}_{\text{res}}$) Algorithmic Representation Leaderboard.

Rank	Method	Accuracy	$b_{\text{eff}}^{\text{agg}}$	VRAM Saved	$\mathcal{S}_{\text{res}}$
1	**KV-Quant**	60.7%	6.04 bpt	62.2%	**61.1**
2	**SnapKV**	58.2%	6.00 bpt	62.5%	59.5
3	**DKV-Mid**	53.3%	5.84 bpt	63.5%	56.3
4	**DKV-High**	59.1%	12.14 bpt	24.1%	48.6
5	**StreamingLLM**	36.5%	6.00 bpt	62.5%	44.3
6	**Dense FP16** (Reference)	60.0%	16.00 bpt	0.0%	42.0
7	**Low-Rank KV**	28.6%	6.09 bpt	62.0%	38.6
8	**KV-Merging**	19.9%	6.03 bpt	62.3%	32.6

Table 6: Qwen2.5 7B Part 2 ($\mathcal{S}_{\text{sys}}$) Hardware System Serving Leaderboard.

Rank	Method	$\mathcal{S}_{\text{sys}}$	$\mathcal{S}_{\text{res}}$	TTFT	Decode	Peak VRAM	ϕ
1	**KV-Quant**	**72.8**	61.1	5.46s	16.4 tok/s	3854.3 MB	1.00×
2	**StreamingLLM**	61.0	44.3	5.51s	24.9 tok/s	3854.4 MB	1.00×
3	**Dense FP16**	59.4	42.0	5.56s	16.3 tok/s	3854.2 MB	1.00×
4	**Low-Rank KV**	54.9	38.6	6.82s	18.1 tok/s	3854.4 MB	0.93×
5	**DKV-Mid**	54.2	56.3	39.55s	15.3 tok/s	3854.2 MB	0.49×
6	**SnapKV**	53.7	59.5	58.69s	22.2 tok/s	5132.6 MB	0.40×
7	**KV-Merging**	52.8	32.6	5.51s	25.8 tok/s	3854.4 MB	1.00×
8	**DKV-High**	48.5	48.6	41.88s	15.3 tok/s	3854.5 MB	0.48×

B Full Task-Wise Breakdown Tables

B.1 Gemma 4 E2B Task Breakdown (5 Tasks × 5 Context Lengths)

Table 7: Gemma 4 E2B Task-Wise Accuracy Breakdown across Context Lengths (2K to 128K tokens).

Task / Method	2K	8K	32K	64K	128K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
KIVI	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
SnapKV	91.7%	100.0%	100.0%	100.0%	100.0%	98.3%
DKV-Mid	100.0%	100.0%	100.0%	100.0%	83.3%	96.7%
Low-Rank KV	100.0%	53.3%	50.0%	50.0%	50.0%	60.7%
StreamingLLM	54.9%	50.0%	50.0%	50.0%	50.0%	51.0%
KV-Merging	48.7%	52.1%	42.7%	52.4%	35.7%	46.3%
Task 2: Multi-Needle Extraction (Multi NIAH)						
DKV-High	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
KIVI	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
Dense FP16	33.3%	33.3%	33.3%	33.3%	27.8%	32.2%
DKV-Mid	33.3%	33.3%	22.2%	11.1%	0.0%	20.0%
Low-Rank KV	33.3%	33.3%	16.7%	5.6%	5.6%	18.9%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
KV-Merging	16.7%	16.7%	16.7%	0.0%	0.0%	10.0%
Task 3: Key-Value Retrieval (RULER KV)						
SnapKV	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
KIVI	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
DKV-High	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
Dense FP16	33.3%	33.3%	33.3%	0.0%	0.0%	**20.0%**
DKV-Mid	33.3%	0.0%	33.3%	0.0%	0.0%	13.3%
Low-Rank KV	33.3%	16.7%	0.0%	0.0%	0.0%	10.0%
KV-Merging	16.7%	0.0%	0.0%	0.0%	0.0%	3.3%
StreamingLLM	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 4: Variable Tracking (RULER Variable Tracking)						
All methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
DKV-Mid	100.0%	100.0%	100.0%	100.0%	66.7%	**93.3%**
SnapKV	100.0%	100.0%	100.0%	100.0%	16.7%	83.3%
KIVI	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
DKV-High	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
Dense FP16	100.0%	100.0%	100.0%	100.0%	0.0%	80.0%
Low-Rank KV	100.0%	100.0%	66.7%	66.7%	0.0%	66.7%
StreamingLLM	83.3%	50.0%	50.0%	50.0%	0.0%	46.7%
KV-Merging	33.3%	0.0%	0.0%	0.0%	0.0%	6.7%

B.2 Gemma 4 E4B Task Breakdown (5 Tasks × 5 Context Lengths)

Table 8: Gemma 4 E4B Task-Wise Accuracy Breakdown across Context Lengths (2K to 32K tokens).

Task / Method	2K	4K	8K	16K	32K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-Mid	100.0%	100.0%	83.3%	83.3%	83.3%	90.0%
KIVI	100.0%	100.0%	100.0%	91.7%	58.3%	90.0%
SnapKV	100.0%	100.0%	100.0%	66.7%	50.0%	83.3%
Low-Rank KV	100.0%	91.7%	60.9%	63.1%	58.8%	74.9%
StreamingLLM	50.0%	52.2%	50.0%	51.1%	50.0%	50.7%
KV-Merging	53.6%	48.5%	38.2%	40.3%	46.6%	45.4%
Task 2: Multi-Needle Extraction (Multi NIAH)						
DKV-Mid	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
KIVI	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
DKV-High	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
Dense FP16	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
Low-Rank KV	33.3%	33.3%	22.2%	16.7%	22.2%	25.6%
KV-Merging	16.7%	16.7%	16.7%	22.2%	33.3%	21.1%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
Task 3: Key-Value Retrieval (RULER KV)						
KIVI	100.0%	100.0%	66.7%	66.7%	100.0%	**86.7%**
Dense FP16	100.0%	100.0%	66.7%	66.7%	100.0%	**86.7%**
SnapKV	66.7%	83.3%	100.0%	83.3%	83.3%	83.3%
DKV-High	100.0%	100.0%	33.3%	33.3%	100.0%	73.3%
Low-Rank KV	100.0%	100.0%	33.3%	33.3%	66.7%	66.7%
DKV-Mid	100.0%	100.0%	33.3%	0.0%	33.3%	53.3%
KV-Merging	33.3%	33.3%	16.7%	16.7%	33.3%	26.7%
StreamingLLM	0.0%	33.3%	0.0%	0.0%	0.0%	6.7%
Task 4: Variable Tracking (RULER Variable Tracking)						
All methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
DKV-Mid	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
SnapKV	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
KIVI	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Low-Rank KV	100.0%	83.3%	66.7%	50.0%	83.3%	76.7%
StreamingLLM	50.0%	66.7%	50.0%	50.0%	50.0%	53.3%
KV-Merging	33.3%	50.0%	16.7%	50.0%	16.7%	33.3%

B.3 Qwen2.5 7B Task Breakdown (5 Tasks × 5 Context Lengths)

Table 9: Qwen2.5 7B Task-Wise Accuracy Breakdown across Context Lengths (2K to 32K tokens).

Task / Method	2K	4K	8K	16K	32K	Mean Acc
Task 1: Single Needle in a Haystack (Single NIAH)						
SnapKV	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
KV-Quant	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	100.0%	**100.0%**
DKV-Mid	100.0%	83.3%	66.7%	47.4%	56.9%	70.9%
StreamingLLM	54.0%	54.3%	53.4%	52.1%	33.1%	49.4%
KV-Merging	22.5%	21.5%	16.8%	14.0%	17.0%	18.4%
Low-Rank KV	31.7%	22.9%	12.8%	5.8%	7.4%	16.1%
Task 2: Multi-Needle Extraction (Multi NIAH)						
Dense FP16	33.3%	33.3%	33.3%	33.3%	33.3%	**33.3%**
SnapKV	27.8%	27.8%	33.3%	33.3%	33.3%	31.1%
KV-Quant	27.8%	33.3%	33.3%	22.2%	33.3%	30.0%
DKV-Mid	22.2%	33.3%	33.3%	22.2%	33.3%	28.9%
DKV-High	22.2%	33.3%	33.3%	33.3%	22.2%	28.9%
StreamingLLM	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%
KV-Merging	5.6%	11.1%	11.1%	5.6%	5.6%	7.8%
Low-Rank KV	11.1%	16.7%	0.0%	5.6%	0.0%	6.7%
Task 3: Key-Value Retrieval (RULER KV)						
KV-Quant	100.0%	100.0%	100.0%	100.0%	50.0%	**90.0%**
DKV-Mid	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
DKV-High	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
Dense FP16	100.0%	100.0%	100.0%	100.0%	33.3%	86.7%
SnapKV	100.0%	100.0%	83.3%	83.3%	33.3%	80.0%
Low-Rank KV	16.7%	83.3%	83.3%	50.0%	33.3%	53.3%
StreamingLLM	50.0%	50.0%	16.7%	33.3%	33.3%	36.7%
KV-Merging	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 4: Variable Tracking (RULER Variable Tracking)						
KV-Quant	0.0%	0.0%	16.7%	0.0%	0.0%	**3.3%**
Other 7 methods	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Task 5: Multi-Hop Reasoning (Multi-Hop QA)						
StreamingLLM	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
SnapKV	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
KV-Quant	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
DKV-Mid	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
DKV-High	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
Dense FP16	100.0%	100.0%	100.0%	100.0%	0.0%	**80.0%**
KV-Merging	100.0%	100.0%	83.3%	83.3%	0.0%	73.3%
Low-Rank KV	100.0%	100.0%	83.3%	50.0%	0.0%	66.7%

C Measurement Protocol & Execution Environment

Hardware Configuration. All evaluations were conducted on dedicated GPU compute nodes equipped with NVIDIA RTX GPUs with 16GB VRAM, CUDA 12.4, and PyTorch 2.4. Model weights were loaded under 4-bit NormalFloat (NF4) quantization via BitsAndBytes to isolate model weight footprint from KV cache dynamics and ensure full evaluation coverage up to 128K context tokens without out-of-memory crashes.

Inference Pipeline. All candidate methods were driven directly from vendored upstream implementations:

- **KIVI (INT4)**: Asymmetric per-channel Key quantization (group size 64) and per-token Value quantization following KIVI (**?**).
- **SnapKV**: Attention pooling kernel $k = 7$ with observation window $L_{\text{obs}} = 32$.
- **DKV (Mid / High)**: Differential KV cache compression with anchor frames and shared low-rank subspace reconstruction (**?**).
- **Low-Rank KV**: Truncated SVD per attention head.
- **StreamingLLM**: 4 initial attention sink tokens with rolling local recency window.
- **KV-Merging**: Temporal adjacent token mean-pooling.

Prefill was executed using chunked forward passes ($C_{\text{chunk}} = 4{,}096$) to avoid PyTorch intermediate tensor allocation spikes. The context memory transform was executed directly on the resident PyTorch KV cache before generating continuation tokens. All decoding used greedy search ($T = 0$) to eliminate sampling variance.

D Proofs and Analysis of Scoring Formulations

Proposition 1 (Monotonicity and Pareto Consistency)**.** Let $\mathcal{S}_{\text{res}}(Q, R) = \alpha Q + (1 - \alpha)R$ with $\alpha \in (0, 1)$. For all $(Q, R) \in [0, 100]^2$, $\mathcal{S}_{\text{res}}$ strictly satisfies Quality Monotonicity, Resource Monotonicity, Boundedness, and Pareto Consistency.

Proof. **1. Monotonicity**: $\frac{\partial \mathcal{S}}{\partial Q} = \alpha > 0$ and $\frac{\partial \mathcal{S}}{\partial R} = 1 - \alpha > 0$.

2. Boundedness: Since $Q, R \in [0, 100]$, the minimum occurs at $(0, 0)$ where $\mathcal{S} = 0$, and the maximum at $(100, 100)$ where $\mathcal{S} = 100\alpha + 100(1 - \alpha) = 100$.

3. Pareto Dominance: If $(Q_A \geq Q_B \wedge R_A \geq R_B)$ with at least one strict inequality, then:

$$\mathcal{S}(Q_A, R_A) - \mathcal{S}(Q_B, R_B) = \alpha(Q_A - Q_B) + (1 - \alpha)(R_A - R_B) > 0. \tag{1}$$

□

E Supplementary Visualizations

Figure 1 illustrates the full distribution of Part 2 hardware system scores across all three foundation models, complementing Tables 2, 4, and 6. Figure 2 provides detailed ranking sensitivity curves across quality preference weight α, and Figure 3 compares candidate utility formulations against the canonical CRBench linear metric.

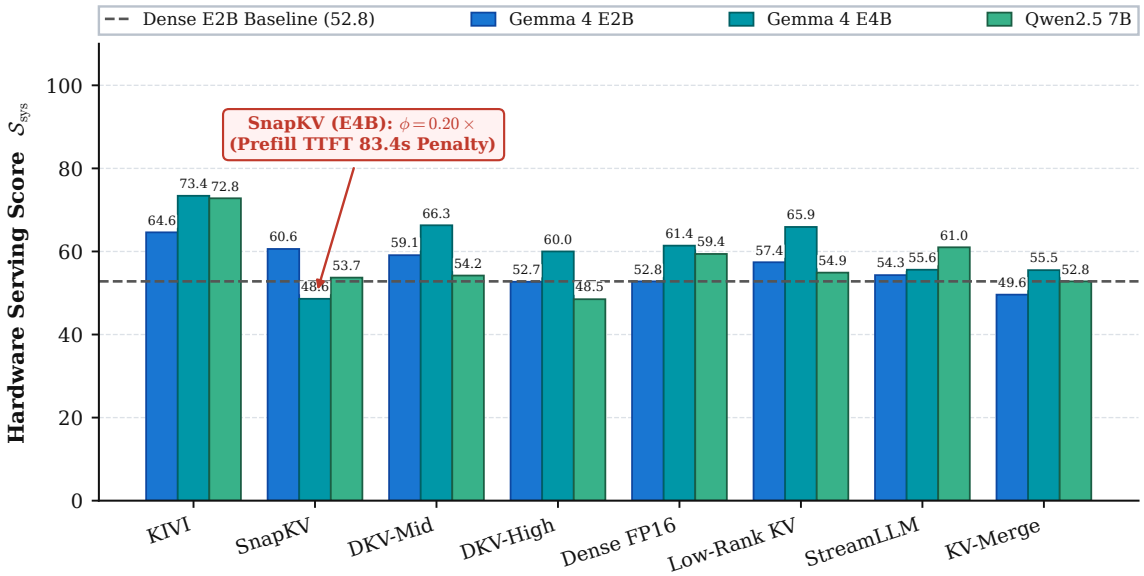


Figure 1: **Cross-Model Part 2 Hardware System Serving Utility Distribution.** Breakdown of $\mathcal{S}_{\text{sys}}$ across Gemma 4 E2B, Gemma 4 E4B, and Qwen2.5 7B.

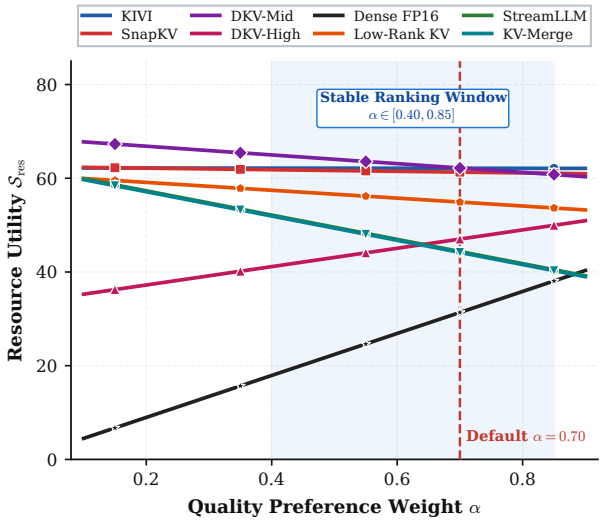


Figure 2: **Ranking Stability Across α.** Trajectories across $\alpha \in [0.1, 0.9]$.

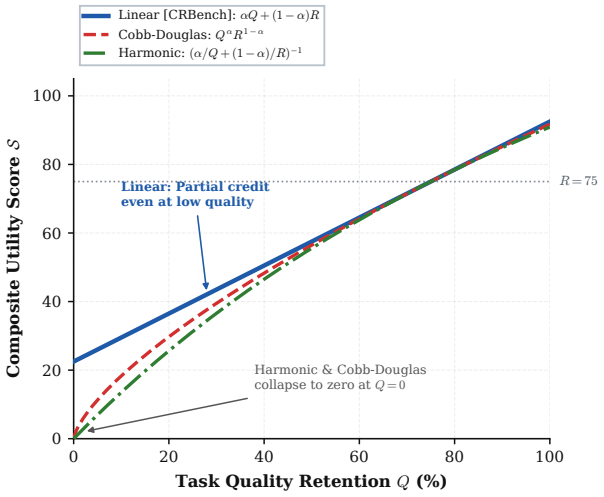


Figure 3: **Utility Formulation Comparison.** Linear vs. Cobb–Douglas vs. Harmonic at $R_{\text{mem}} = 75\%$.